ALARD UNIVERSITY

DSA MINI PROJECT REPORT

STUDENT RECORD MANAGEMENT SYSTEM

Using C++ and Data Structures

Project Details

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Concept	Use in Project	Typical Complexity
Structure	Groups roll number, name, and marks into one record.	—
Array	Stores multiple student records.	Access: O(1)

Concept	Use in Project	Typical Complexity
Linear Search	Finds a record by checking roll numbers one by one.	O(n)
Bubble Sort	Sorts records by roll number.	$O(n^2)$
Functions	Keeps program operations modular and readable.	—

Test	Input / Action	Expected Result	Status
1	Add a valid student	Record is added	Pass
2	Display when records exist	All stored records appear	Pass
3	Search an existing roll number	Matching record appears	Pass
4	Search a missing roll number	Not-found message appears	Pass
5	Sort multiple records	Roll numbers appear ascending	Pass
6	Enter marks outside 0-100	Record is rejected	Pass